CODEALPHA

ROBOTICS & AUTOMATION INTERNSHIP

TASK 3 — AUTOMATION SYSTEM DESIGN

Automated Smart Home Lighting System

Sensor-Based Automatic Lighting with Manual Override

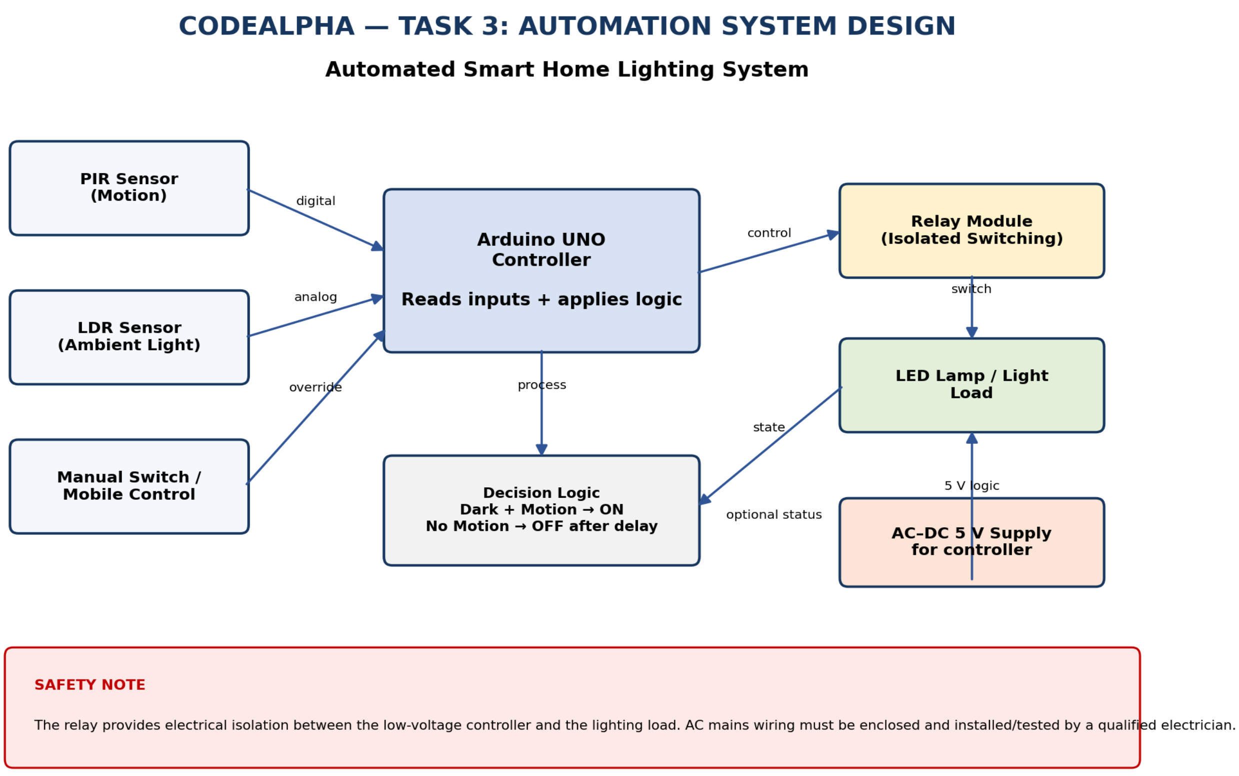

Figure 1. System architecture and control flow for the proposed smart lighting automation system.

Objective: Automatically switch lights according to ambient light and human presence while retaining manual control.

Submission deliverables: Design diagram + short explanation report (2–3 pages), matching the CodeAlpha Task 3 requirement for an automated system such as a smart home lighting system.

1. System Overview

The Automated Smart Home Lighting System is designed to turn lights on only when they are needed. An LDR (Light Dependent Resistor) measures ambient brightness, while a PIR (Passive Infrared) sensor detects human movement. An Arduino UNO acts as the controller and evaluates both inputs before commanding a relay module. The relay switches the lighting load while maintaining separation between the low-voltage control circuit and the mains load.

The system can be used for a living room, bedroom, corridor, staircase, balcony, office, or other indoor space. A manual wall switch or mobile control can be added as an override so that the user can force the light ON or OFF when required.

2. Main Components

Component	Function	Interface / role
Arduino UNO	Central controller	Reads sensors and controls relay
PIR sensor	Detects human motion	Digital input
LDR + resistor	Measures ambient light	Analog input
1-channel relay module	Switches lighting load	Digital output; isolated switching
LED lamp	Lighting load	Controlled output
AC–DC 5 V supply	Powers low-voltage electronics	Controller/sensor supply
Wall switch / mobile control	Manual override	User input

Table 1. Major hardware and their roles.

3. Operating Logic

1. The controller starts and initialises the PIR sensor, LDR input, relay output, and any manual-control input.
2. The LDR is read continuously. If the environment is sufficiently bright, the light remains OFF unless the manual override is used.
3. If the environment is dark, the controller checks the PIR sensor for motion.
4. When motion is detected in a dark environment, the relay is activated and the light turns ON.
5. If motion stops, a preset delay can be used before switching the light OFF. This prevents rapid ON/OFF switching.
6. The cycle repeats continuously while manual control remains available.

4. Control Flow

The control strategy combines two conditions: **brightness** and **presence**. The AND-style decision prevents the light from operating unnecessarily during daytime or when a space is empty.

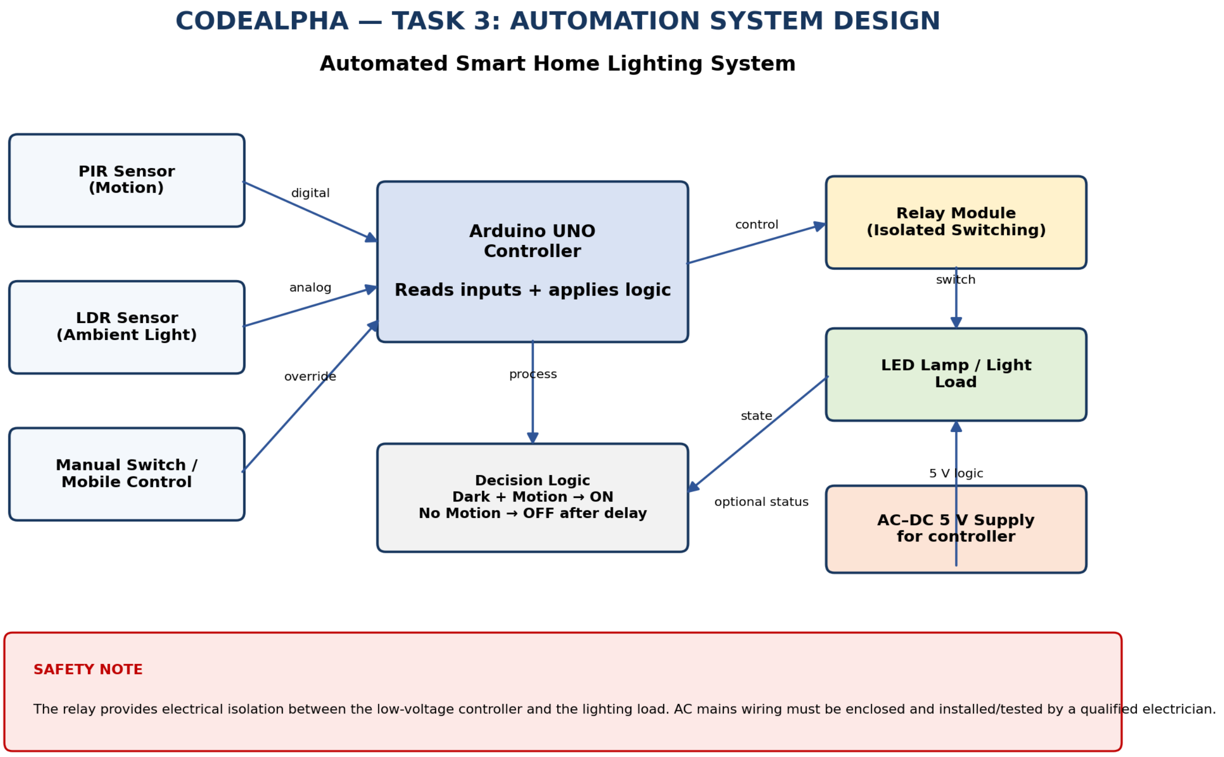

Figure 2. Detailed control architecture. For implementation, the Arduino decision logic can be represented as: if (dark AND motion) → light ON; otherwise → light OFF after the configured delay, subject to manual override.

5. Advantages

- **Energy efficiency:** Lights operate primarily when darkness and occupancy are detected.
- **Convenience:** Automatic operation reduces the need to operate switches manually.
- **Scalability:** Additional sensor-relay pairs can be added for multiple rooms.
- **Low-cost prototype:** The controller and common sensors are widely available for educational projects.
- **Manual override:** Users retain direct control when automation is not desired.

6. Applications

The design is suitable for homes, corridors, staircases, offices, classrooms, storage areas, washrooms, and other spaces where occupancy-based lighting can reduce unnecessary energy use.

7. Safety and Implementation Notes

The Arduino and sensors operate at low voltage, while household lighting may use hazardous AC mains voltage. The relay module must be correctly rated for the lamp load and installed inside a suitable enclosure. The low-voltage circuit should remain isolated from mains wiring. AC wiring, testing, and final installation should be performed by a qualified electrician. For a student prototype, a low-voltage lamp or certified isolated training load is preferable.

8. Conclusion

The proposed smart home lighting system demonstrates a practical automation architecture using sensors, a microcontroller, decision logic, and a relay-controlled output. It satisfies the CodeAlpha Task 3 concept of designing an automated system and provides a clear foundation for future IoT features such as Wi-Fi monitoring, mobile-app control, scheduling, and energy-use logging.

Design File

The engineering diagram is supplied separately as a PNG design file and is also embedded in this report.